Key Formulas

Sample Mean

$$\bar{x} = \frac{\Sigma x_i}{n}$$

Population Mean

$$\mu = \frac{\Sigma x_i}{N}$$

Interquartile Range

$$\text{IQR} = Q_3 - Q_1$$

Population Variance

$$\sigma^2 = \frac{\Sigma(x_i - \mu)^2}{N}$$

Sample Variance

$$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1}$$

Standard Deviation

$$\text{Sample standard deviation} = s = \sqrt{s^2}$$

$$\text{Population standard deviation} = \sigma = \sqrt{\sigma^2}$$

Coefficient of Variation

$$\left(\frac{\text{Standard deviation}}{\text{Mean}} \times 100\right)\%$$

Sample Covariance

$$s_{xy} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Population Covariance

$$\sigma_{xy} = \frac{\Sigma(x_i - \mu_x)(y_i - \mu_y)}{N}$$

Counting Rule for Combinations

$$C_n^N = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

Counting Rule for Permutations

$$P_n^N = n!\binom{N}{n} = \frac{N!}{(N-n)!}$$

Computing Probability Using the Complement

$$P(A) = 1 - P(A^c)$$

Addition Law

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication Law

$$P(A \cap B) = P(B)P(A \mid B)$$

$$P(A \cap B) = P(A)P(B \mid A)$$

Multiplication Law for Independent Events

$$P(A \cap B) = P(A)P(B)$$

Bayes' Theorem

$$P(A_i \mid B) = \frac{P(A_i)P(B \mid A_i)}{P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2) + \cdots + P(A_n)P(B \mid A_n)}$$

Discrete Uniform Probability Function

$$f(x) = 1/n$$

Expected Value of a Discrete Random Variable

$$E(x) = \mu = \Sigma x f(x)$$

Variance of a Discrete Random Variable

$$\mathrm{Var}(x) = \sigma^2 = \Sigma (x - \mu)^2 f(x)$$

Number of Experimental Outcomes Providing Exactly x Successes in n Trials

$$\binom{n}{x} = \frac{n!}{x!(n - x)!}$$

Binomial Probability Function

$$f(x) = \binom{n}{x} p^x (1 - p)^{(n-x)}$$

Expected Value for the Binomial Distribution

$$E(x) = \mu = np$$

Variance for the Binomial Distribution

$$\mathrm{Var}(x) = \sigma^2 = np(1 - p)$$

Poisson Probability Function

$$f(x) = \frac{\mu^x e^{-\mu}}{x!}$$

Normal Probability Density Function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

Converting to the Standard Normal Random Variable

$$z = \frac{x - \mu}{\sigma}$$